Whitney Bullock

Jordan Sharples

Sina Odejinmi

MANUFACTURE FOR THE FUTURE

MX Conversion Optimization

Mastering the transition from initial introduction to long-term substantial conversion through data analysis and ideal customer profiling.

MSBA University of Utah
Capstone Presentation

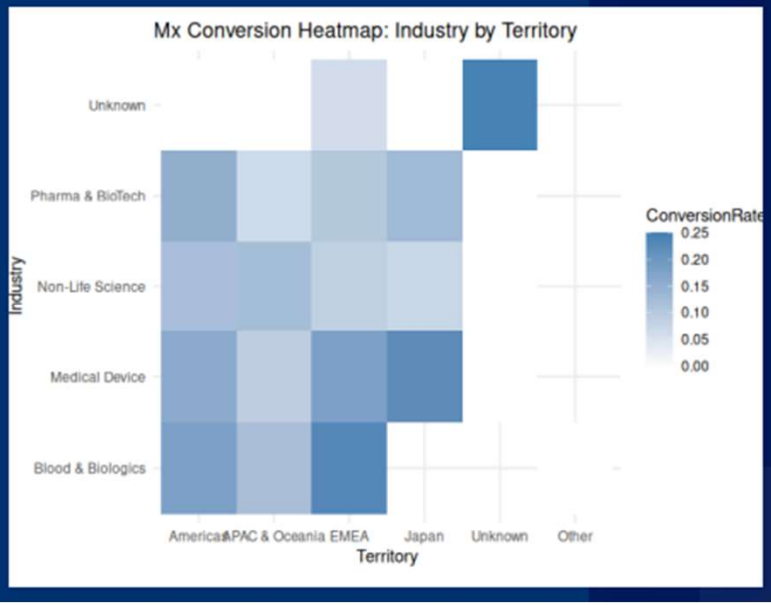
Primary Objectives

Identify which **industries, manufacturing models, site functions, and company sizes** are best suited

Which **job titles, seniority levels, and functional roles** are most influential in driving sales progression

Which existing Qx customers are most likely to be **adopters of Mx** as an additional solution in their portfolio

EDA



MX 12.7%

Qx 19%

Higher Conversions

S/M Size Companies **17%**
Blood & Biologics / Medical Devices **17%**
Americas / EMEA **15-20%**
Consumer Packaged Goods, OEM, CMO **18-22%**
High-Intent, Contact Us, Demo **20%**

EMEA x Blood & Biologics = 25% !?!?

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MX Conversion Rate

MasterControl

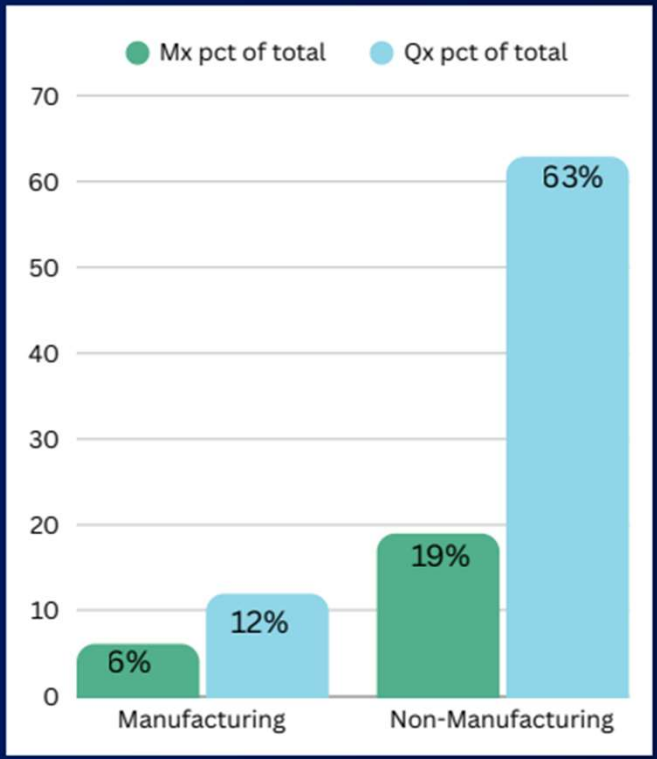
IDEAL CUSTOMER PROFILE IDENTIFIED

Comparison Mx vs Qx Performance

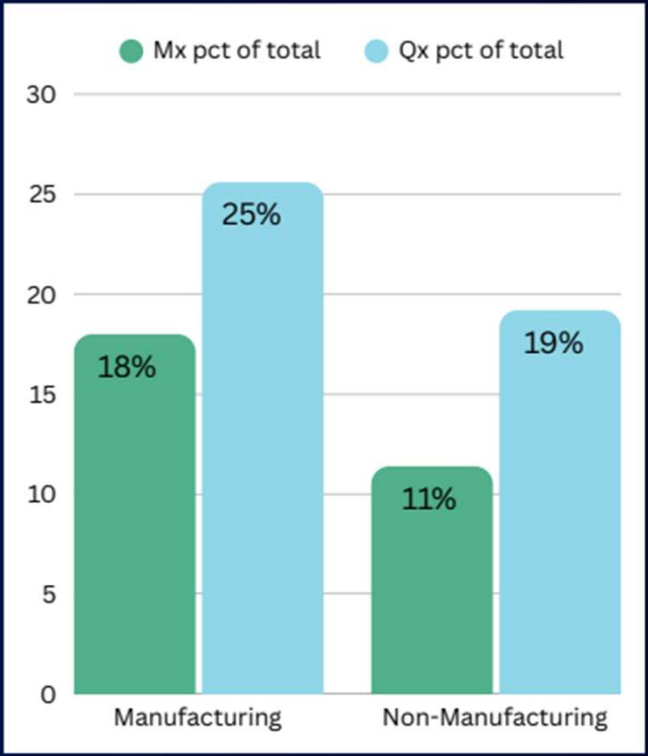
Qx dominates total lead volume

Qx also converts better

Fewer Manufacturing leads but convert at higher rates for both solutions



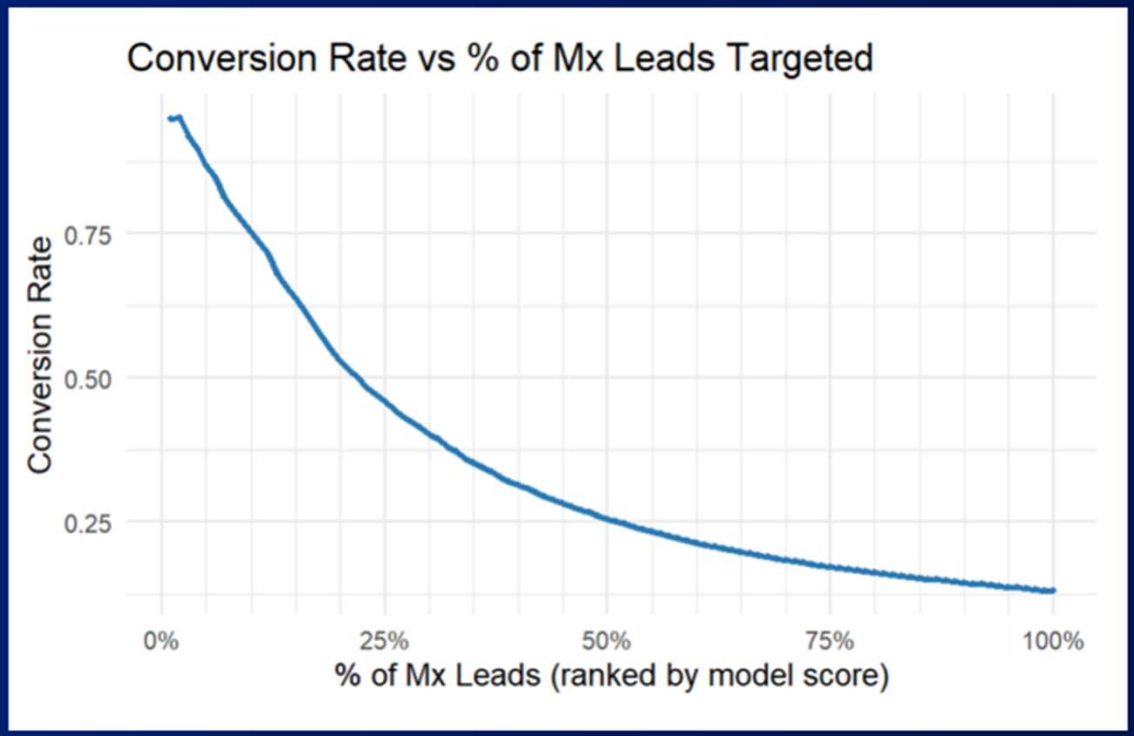
Proportion of Mx to Qx Leads



Conversion Rate Mx and Qx

Modeling AUC

- Logistics Regression 0.8165
- Random Forest 0.8364
- Gradient Boosted Model 0.8291



Model is highly effective at ranking Mx leads

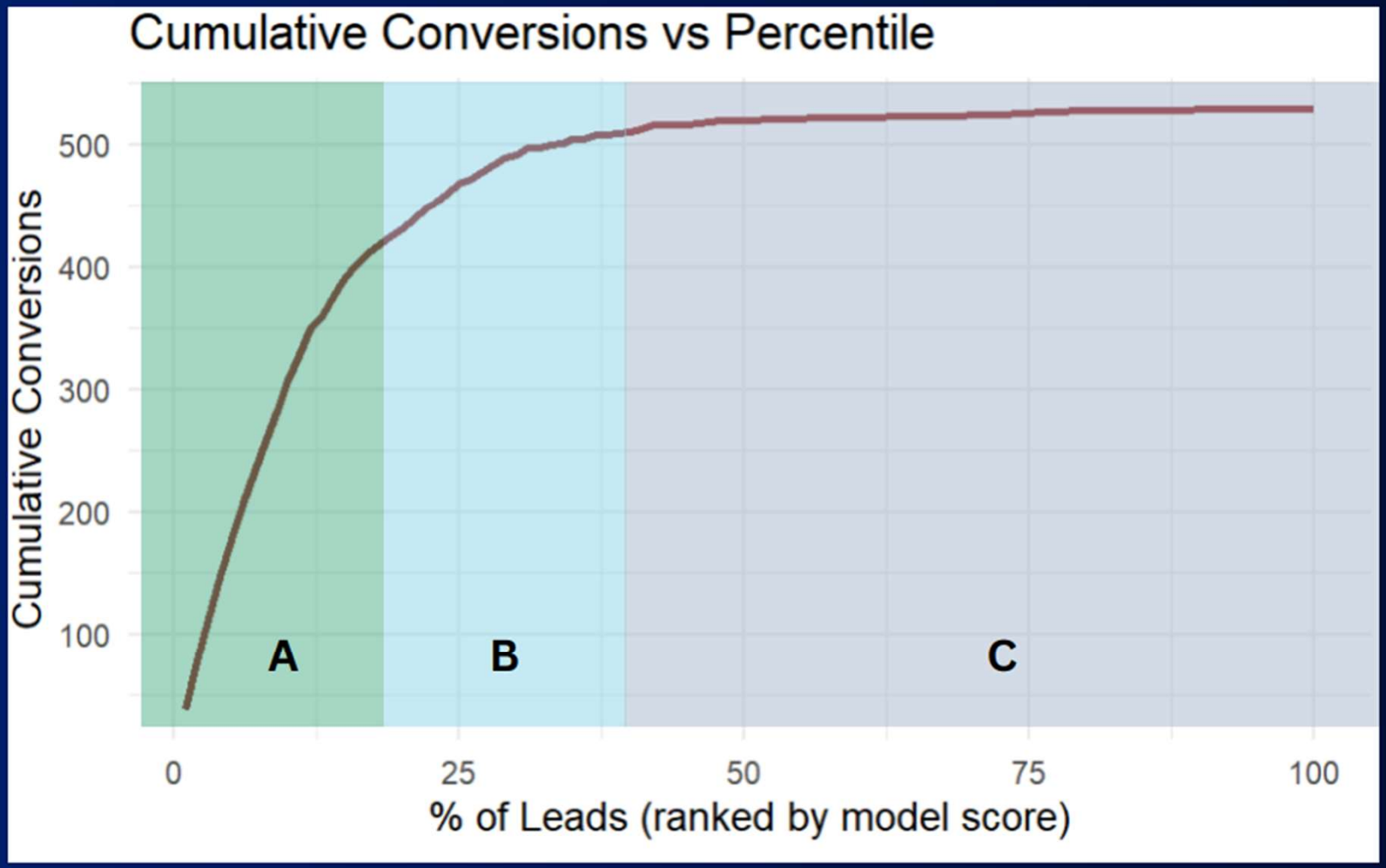
High-scoring leads = higher conversion rates

Lower-scoring leads, performance declines

Conversion Rates by Percentile (1-25)

percentile <dbl>	conversion_rate <dbl>
1	0.95
5	0.87
10	0.75
20	0.53
25	0.46

Most Conversions Are Driven by the Top Percentiles



Model's Generalizability Across Segments

Percentile	total	progressed	conversion_rate
100%	97	13	13.4%
50%	27	6	22.2%
20%	9	5	55.6%

Table 1: CDMO-Pharma & BioTech

Percentile	total	progressed	conversion_rate
100%	48	9	18.8%
50%	27	9	33.3%
25%	10	6	60.0%

Table 2: In-House-Quality/Regulatory

High Probability of Converting Prediction

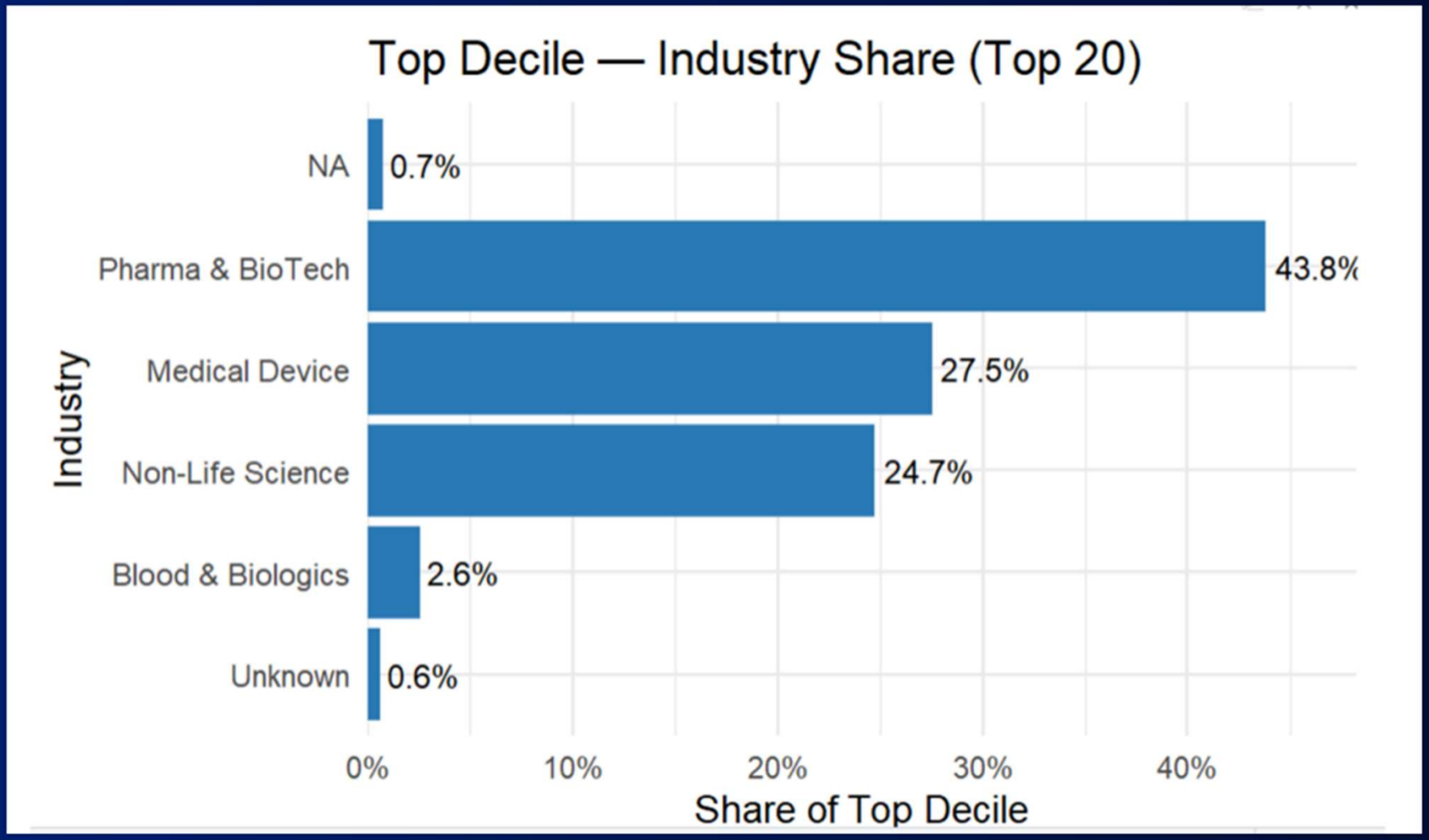
.pred_yes = **71%**
.pred_no = **29%**

ICP Dimension	This Lead	ICP Fit
Industry	Medical Device	✔ Strong
Manufacturing Model	OEM	✔ Strong
Site Function	Diagnostic Equipment	✔ Strong
Title	Quality/Compliance Manager	✔ Strong
Region	EMEA	✔ Good
Tier	Small	✔ Acceptable
Solution	Mx	✔ Perfect
Model Score	Top 10%	✔ Very Strong

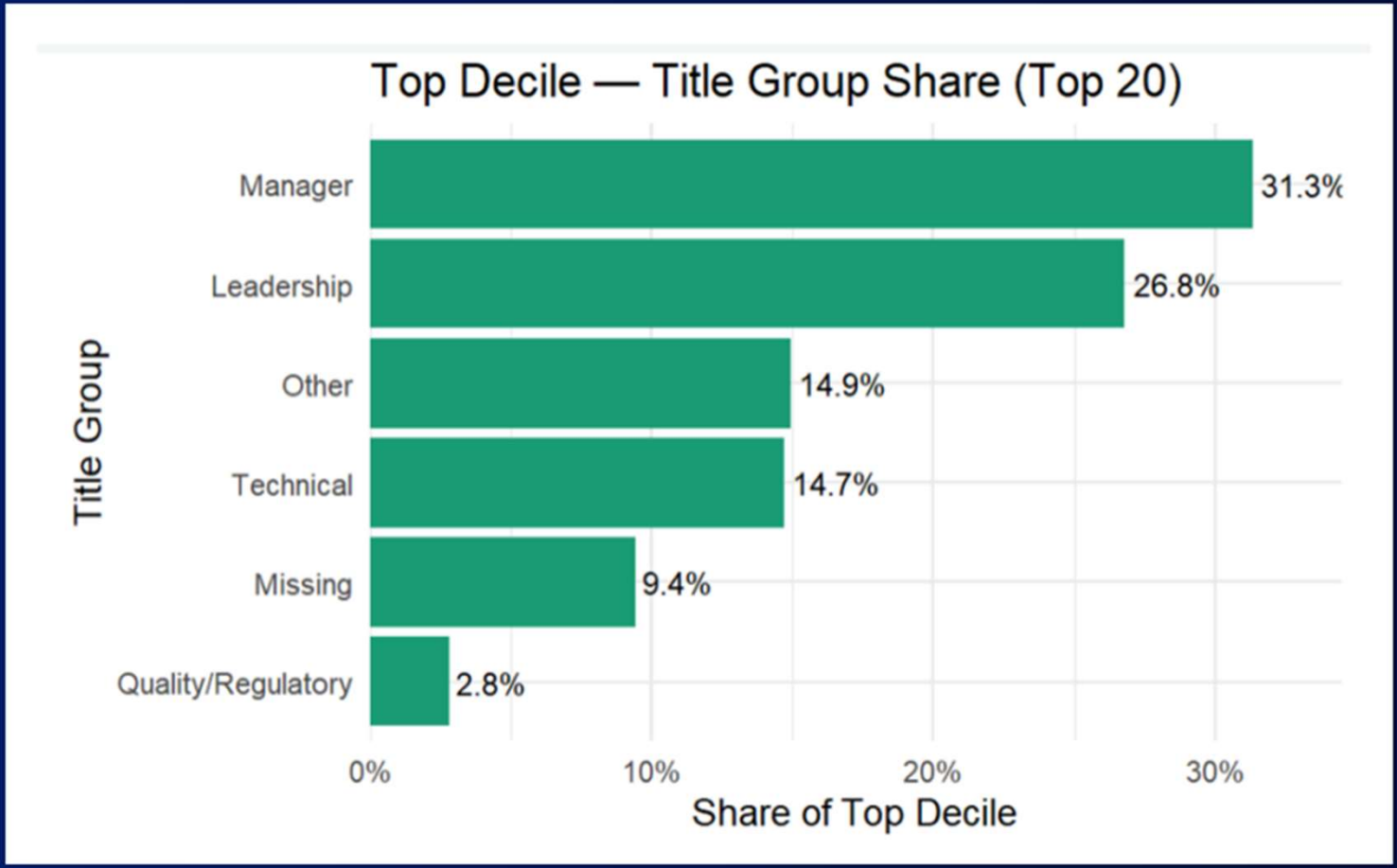
Low Intent Converting Prediction

.pred_yes = **38%**
.pred_no = **62%**

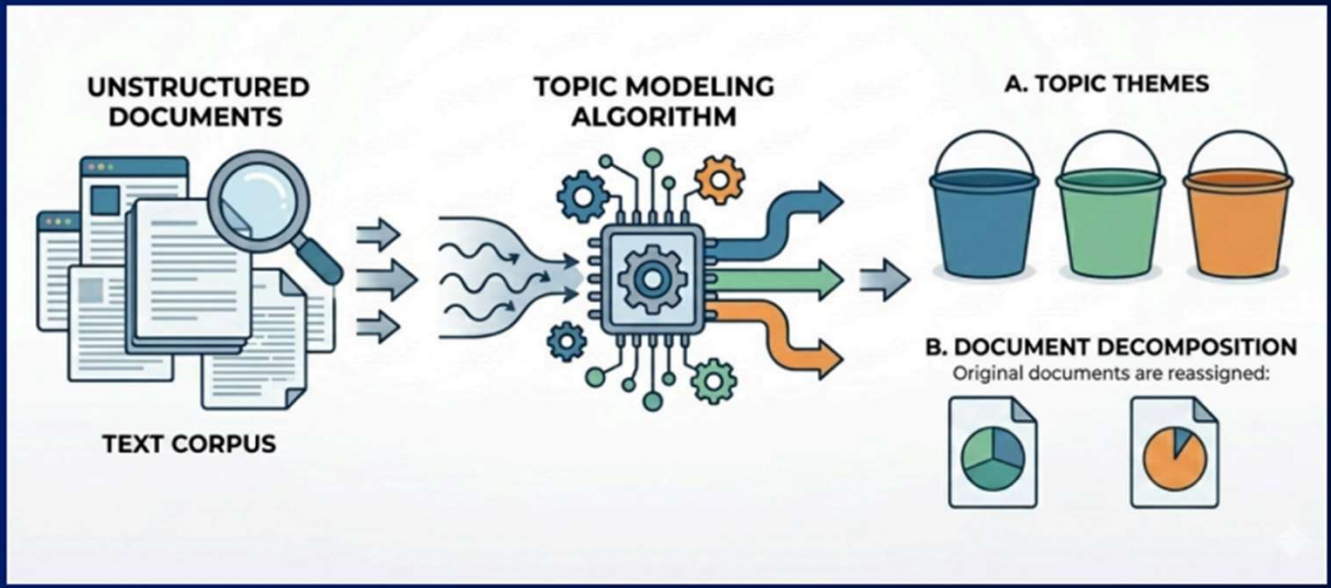
ICP Dimension	Lead Value	ICP Fit
Industry	Retail	❌ Not ICP
Manufacturing Model	Outsourced	❌ Not ICP
Site Function	Corporate Services	❌ Not ICP
Title	Administrative Assistant	❌ Not ICP
Region	APAC	⚠️ Neutral
Tier	Small	⚠️ Neutral
Solution	Mx	⚠️ Partial
Intent Signals	Low	❌ Not ICP


Top Decile — Industry Share (Top 20)
Industry
NA
Pharma & BioTech
Medical Device
Non-Life Science
Blood & Biologics
Unknown
0.7%
43.8%
27.5%
24.7%
2.6%
0.6%
0%
10%
20%
30%
40%
Share of Top Decile

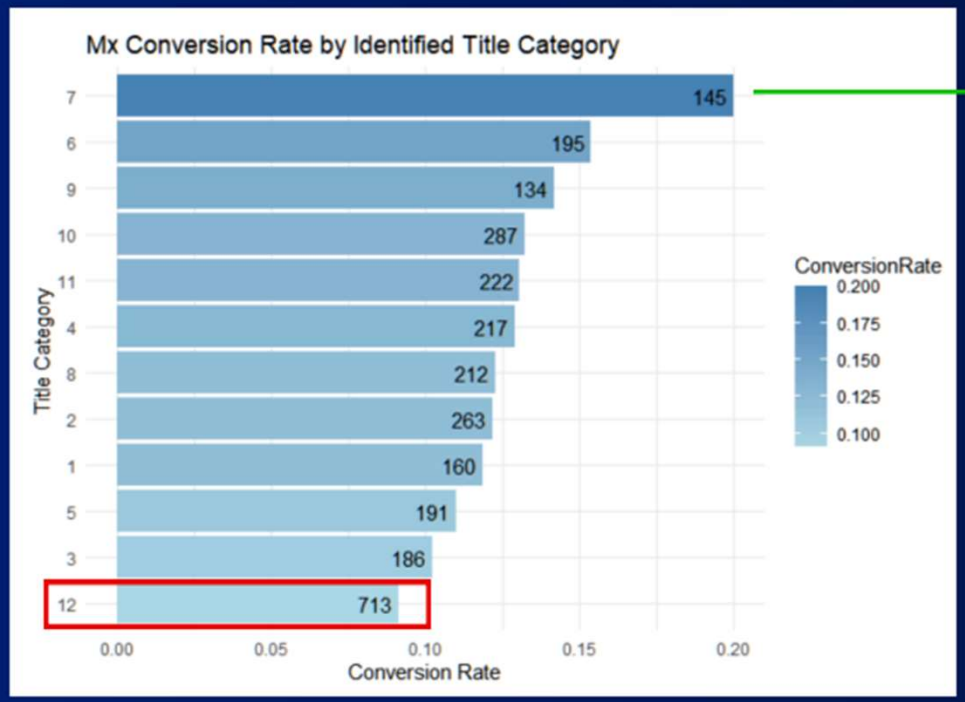
Targeted Outreach


Top Decile — Title Group Share (Top 20)
Title Group
Manager
Leadership
Other
Technical
Missing
Quality/Regulatory
31.3%
26.8%
14.9%
14.7%
9.4%
2.8%
0%
10%
20%
30%
Share of Top Decile

What is Topic Modeling?



Contact Title Topic



Title Group Top Cross Performance

Site Function

- Group 7 + Other Manufacturing - 26%
- Group 12 + Device Manufacturing - 25%

Industry

- Group 7 + Medical Device - 24%
- Group 8 + Medical Device - 22%

Manufacturing Model

- Group 10 + CMO - 24%
- Group 8 + CDMO - 23%

Company size

- Group 6 + Small - 33%
- Group 7 + Small - 32%

Objectives Achieved

Ideal Customer Profile

Ranked list of high predicting conversions

Targeted Outreach

Identification of key decision makers

Opportunity to surpass target benchmark of 19%

Inquiries

- Whitney Bullock
- Jordan Sharples
- Sina Odejinmi

Project Github